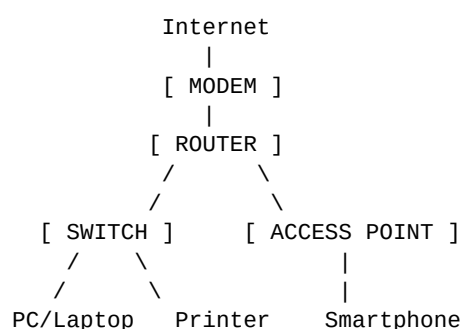


Day 1 – Introduction to Networking Hardware

Networking Hardware Assignment

Task 1: Home Wi-Fi Network Diagram



Simple Explanation

- **Modem:** Connects the home network to the Internet service provider.
- **Router:** Shares Internet and manages traffic between networks.
- **Switch:** Connects multiple wired devices in the local network.
- **Access Point:** Provides Wi-Fi connection to wireless devices.
- **Laptop / Smartphone:** Devices that use the network.

Task 2: Main Function of Networking Hardware

Device	Main Function	Real-life Example
Modem	Connects the home network to the ISP/Internet connection.	Used at home with broadband.
Router	Shares Internet and directs data between networks.	Used at home to share Wi-Fi.
Switch	Connects multiple wired devices in the same LAN.	Used in an office for PCs/printers.
Access Point	Provides Wi-Fi access to network devices.	Used in an office, school or hotel.
Network Cable	Carries data between networking devices.	Ethernet cable from PC to router.

Task 3: Types of Network Cables

1. Ethernet Cable

Cat5e / Cat6 cable. Used for connecting computers, routers, switches and other network devices in LAN networks.

PC ----- Ethernet Cable ----- Switch

2. Coaxial Cable

Used for cable TV, broadband Internet and some older networking systems.

Cable Line ----- Coaxial Cable ----- Modem

3. Fiber Optic Cable

Used for high-speed Internet and long-distance communication using light signals.

ISP ----- Fiber Optic Cable ----- Network Equipment

Image Sources

Ethernet cable – Wikimedia Commons: https://commons.wikimedia.org/wiki/Category:Ethernet_cables

Coaxial cable – Wikimedia Commons: https://commons.wikimedia.org/wiki/Category:Coaxial_cables

Task 4: Router vs Switch Comparison

Difference	Router	Switch
Main Use	Connects different networks and provides Internet access.	Connects devices within the same LAN.
Typical Location	Homes, offices and Internet gateways.	Offices, computer labs and data networks.
Data Handling	Uses IP addresses to route packets between networks.	Mainly uses MAC addresses to forward frames within a LAN.
Internet Sharing	Can share one Internet connection with many devices.	Does not normally provide Internet access by itself.
Wireless Connection	Many home routers provide Wi-Fi.	Basic switches normally provide wired connections only.
Example	Home Wi-Fi router.	Office network switch.

Conclusion: Networking hardware helps devices communicate. A modem connects to the Internet, a router manages network traffic, a switch connects wired devices, an access point provides Wi-Fi, and cables carry data.